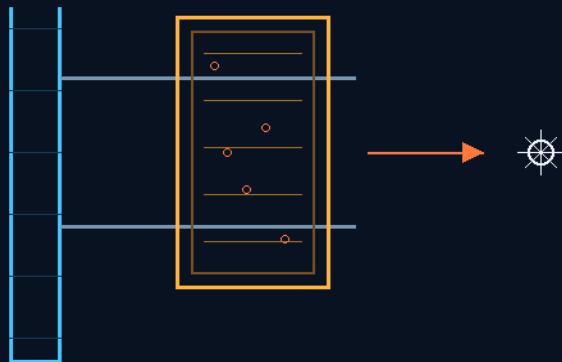


THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS



PHASE 12

AGAR-PLATED EXOBIOLOGICAL BIO-SIPHONS & LEVEL-5 INCINERATION LOCKS

The barrel points outward: nothing off-world
comes aboard unplated, unscreened, unanswered.
Deploy, expose, retract, seal — the weekly loop.
Uncontainable: the module leaves, toward the star.

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THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 12

PHASE 12: AGAR-PLATED EXOBIOLOGICAL BIO-SIPHONS & LEVEL-5 INCINERATION LOCKS — STATUS: CURRENT. Automated biological safety barriers across all hangar entries — real-time agar testing plates and emergency multi-point high-temperature incineration valves to neutralize unknown exobiological contamination vectors. **The index law, verbatim: "All areas have them. Planetary and water modules also."** The doctrine, seated in the ship registry: anything biological from off-world passes through agar-plated barriers and Level-5 burn locks. No exceptions, ever. (The render is APPROVED — sitting 28, the Author's verdict: "2 were good" — Phases 11 and 12, final.)

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat 3 AI (The Initial Interstellar Co-Engineering Alliance). License: CERN Open Hardware Licence v2 (CERN-OHL-W-2.0). Manual GP-IM-012 v1.0 — 19 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the doctrine — §2 the bio-siphon panels — §3 the level-5 incineration locks — §4 the agar supply chain — §5 where the barriers stand — §6 build sequence — §7 cross-phase — §8 do-not — §9 owed — §10 provenance & change control. **THE MASTER WINS:** where this manual and the master ever differ, the master is law and this manual is reprinted.

§1 — THE DOCTRINE (THE BARREL POINTS OUTWARD)

Phase 12 is the ship's immune border. Its barrel points OUTWARD: it quarantines off-world material at the boundary — hangar entries, sample returns, planetary and water modules — catching unknown biology on food it cannot refuse (nutrient agar), reading what grew, and burning what cannot be contained. The seated vet cross-check grades the coverage honestly: covered at the border — and the inward question (the ship's own water loops, cabin air, the crew's biome) is flagged amendment-class, carried at §9 as the Author's call.

§2 — THE BIO-SIPHON PANELS (THE WEEKLY LOOP)

- THE PANEL: multi-metre hull panels coated in vacuum-stable nutrient agar gel. The gel face is the trap — a broad biological assay plate riding the hull itself.
- THE CYCLE: panels slide out into the free-floating drift matrix of planetary debris — deploy, expose, retract, seal — then retract WEEKLY through vacuum locks into automated incubation locks, screening for cellular proteins. Whatever the drift is carrying, the agar catches it and the incubation lock names it before it ever meets a crewmember.
- THE RENDER OF RECORD: sitting 28 — 'Phase 12 — The Agar Bio-Siphon Panel.mp4', prompt archived per the Prompt-Archive Law, audit frame-read LANDED (the full weekly cycle in one shot: panel out on rails, glistening translucent amber in the starlight, home through the slot, heavy door sealing flush). The Author's verdict: APPROVED, final.

§3 — THE LEVEL-5 INCINERATION LOCKS (THE FINAL ANSWER)

- THE TRIGGER: an uncontrollable level-5 outbreak in the screening chain — the incubation lock reads something the barrier cannot hold.
- THE ACTION: immediate symmetric deadbolt uncoupling — the locks part from both sides at once — and the ENTIRE lab module is ejected toward the nearest star for total atomic incineration. No partial measures, no crew exposure, no debate: the module leaves, and the star does the sterilizing.
- THE CONSISTENCY: this is the same ejection doctrine as the sitting-475 bio/chem pod law (Phase 6 — bio pods and chemistry pods always outer wall; contamination is simply ejected toward the sun/a star; the hull pod is replaced). Phase 12 wrote the doctrine first; Phase 6 built it into the matrix. Emergency multi-point high-temperature incineration valves stand at the hangar entries as the in-place burn layer under the same law.

§4 — THE AGAR SUPPLY CHAIN (GROWN, NOT SHIPPED)

- THE STOCK: raw seaweed (red macroalgae) and algae biomass cultivated in the lower-deck botanical greenhouses.
- THE PURIFICATION: thermal boiling and a simple freeze-thaw phase-change process purify raw agar-agar — no chemical reagents.
- THE DOUBLE DUTY: the greenhouse stock acts as a continuous photosynthetic CO2 scrubber while it grows the barrier's plates. The ship grows its own immune system and breathes easier for it.

§5 — WHERE THE BARRIERS STAND

The index law: ALL areas — every hangar entry — and the planetary and water modules also. Sample-return routing carries untainted samples to the Phase 12 exobiological chambers for level-5 isolation dissection by grained robotic labor (Phase 49), the sealed cassette routed by the Phase 20 tether reclamation line into the secondary sub-deck hangar bays.

§6 — BUILD SEQUENCE

- 1. Fit the panel rails and vacuum locks at every hangar entry; plate the panels with vacuum-stable nutrient agar.
- 2. Build the automated incubation locks behind the locks — weekly retract, screen for cellular proteins.
- 3. Fit the emergency multi-point high-temp incineration valves at the entries; wire the symmetric deadbolt uncoupling on the lab module.
- 4. Stand up the agar supply: greenhouse macroalgae stock, boil + freeze-thaw purification line, plate-casting for the panels.
- 5. Extend the coverage to the planetary and water modules per the index law. The inward water-loop question is §9 — the Author's call.

§7 — CROSS-PHASE (POINTERS)

- PHASE 10 — the UV laser family sterilizes the corridor bio-gel cassettes (the Phase 24 ribbon array, same cluster): scanned, sterilized, re-melted, re-poured in a material-free closed loop. The cassette assay, clarified by the Author 12 August: collect daily, replace, growth cabinet ~3 days, observe growth — the colour-change reading was an auditor's invention, withdrawn.
- PHASE 13 — the AION plating the planetary seed pods wear (the cluster's member 25 rides this plating).
- PHASE 6 — the sliding matrix and the bio/chem outer-wall ejection law (GP-IM-006 v1.2) — the same doctrine, built into the pod grid.
- PHASES 24, 25, 26 — the cluster-mates: hangar scrub boundaries & corridor cassettes; mycorrhizal seeding pods; the Snowglobe hydroponics. This phase shares its combined cluster header with them pending any split order.
- PHASES 20 & 49 — tether reclamation routing of the sealed cassette; grained robotic dissection labor in the level-5 chambers.

§8 — DO NOT

- DO NOT waive the barrier — no exceptions, ever; the registry line is law, and 'all areas have them' means all.
- DO NOT improvise a partial response to a level-5 trigger — the answer is whole-module ejection starward; anything less is exposure.
- DO NOT read colour off a cassette or plate as the assay — the clarified assay is collect, incubate ~3 days, observe growth; the colour-change reading was the external auditor's own invention, withdrawn and seated as such.
- DO NOT ship agar from Earth as the plan of record — the supply chain is grown aboard (§4); stores are bootstrap, not doctrine.

§9 — OWED (THE AUTHOR'S CALL)

- THE INWARD EXTENSION — the seated vet cross-check, verbatim in spirit: covered at the border, open inside the hull. The ship's own water loops, cabin air, surfaces, and the crew biome fight the same biofilm the ISS fights; the candidate amendment is to extend Phase 12's plate-and-incubation doctrine inward to water-loop coupons. Amendment-class, not phase-class — the Author's call, flagged and waiting.
- THE ASSAY SPEC — sensitivity and target organisms for the plate/cassette screening: owed, and per the Phase 11 ruling pattern it closes as a bench item unless he speaks it first.

§10 — STATUS, PROVENANCE & CHANGE CONTROL

STATUS: CURRENT — GP-IM-012, v1.0, 19 August 2026. Twelfth phase of 290; thirteenth manual delivered. The border is law, the loop is weekly, the star is the incinerator.

PROVENANCE: built from the master's 12/24/25/26 cluster (member-12 scope: the bio-siphon panels, the level-5 incineration locks, the agar synthesis supply chain), the index law (all areas; planetary and water modules also), the ship-registry doctrine line, the sitting-28 render archive (prompt seated, outcome LANDED, his approval final), the 12 August cassette-assay clarification, and the seated vet cross-check with its inward-gap candidate. **THE MASTER WINS.** Where this manual and the master ever differ, the master is law and this manual is reprinted.

CHANGE CONTROL: v1.0 (19 August 2026, sitting 486) — first issue, built on the Author's order, verbatim: "12".

END OF MANUAL — PHASE 12